



AI & Data COE Spotlight Session

“AI & Data empower modernization, disrupt boundaries, and amplify value creation. Anchored by SPEED capabilities and a brain empowered by data, we deliver 10x exponential solutions to redefine the future.”

Real-Time Data Processing on Databricks: From Streaming Events to Contextual AI Insights

Agenda

POC Context and Problem Statement

What We Built: Real-Time Data and AI Pattern on Databricks

Databricks Services and Supporting Components Used

High-Level Architecture

End-to-End Data Flow

AI-Enabled Insight Generation

Live Demo

Reusability Across Future Projects

Lessons Learned and Future Enhancements

POC Context: Real-Time Data to AI-Driven Insights

In our POC, we are simulating a real-time data scenario where continuous events (such as conversations or interactions) are ingested, processed, and enriched within Databricks.

The objective is to demonstrate how raw streaming data can be transformed into structured, context-aware information and further leveraged using Generative AI to provide real-time insights and assistance.

This POC focuses on combining:

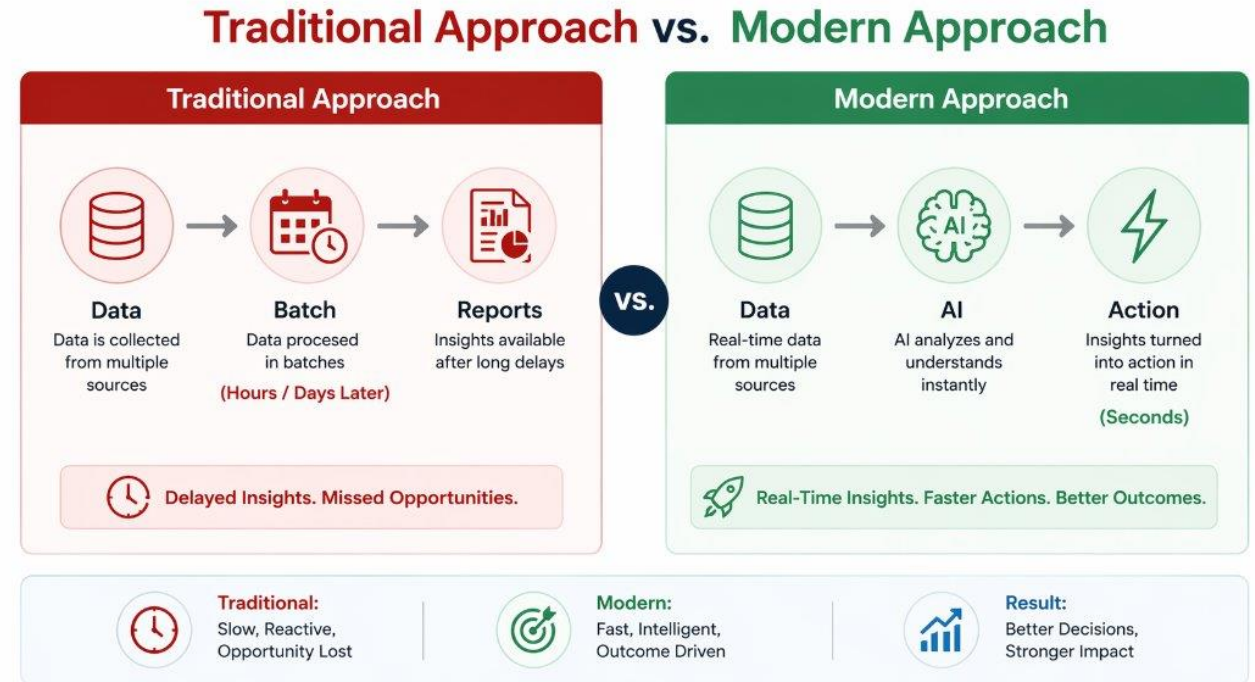
- Real-time data ingestion and processing
- Contextual data enrichment
- AI-based insight generation

From Delayed Insights to Real-Time Decisioning



Problem Statement: Gap Between Data and Real-Time Insights

- Traditional data pipelines are batch-driven, leading to delays between data generation and insight availability
- By the time insights are generated, the opportunity to act in real time is often lost
- Systems lack the ability to interpret data during live interactions and provide contextual guidance
- Decision-making remains reactive, relying on historical reports rather than current events
- There is a need to shift from delayed analytics to real-time, context-aware intelligence



Real-Time Data and AI Pattern on Databricks

A reusable architecture for real-time data processing, enrichment, and AI-driven insights

We have implemented a reusable real-time data and AI pattern on Databricks that enables continuous ingestion, processing, and contextual enrichment of streaming data.

This pattern demonstrates how raw events can be transformed into structured intelligence and leveraged using Generative AI to provide real-time insights and assistance.

Key capabilities include:

- Continuous real-time ingestion using Zerobus
- Streaming data processing using layered architecture (Bronze, Silver, Gold)
- Contextual enrichment (sentiment, intent, entities, compliance signals)
- AI-powered insights and recommendations using LLM and retrieval
- Real-time alerts and operational visibility

System Capabilities:

- ✓ Live transcript ingestion (sub-second latency)
- ✓ Real-time sentiment and intent detection
- ✓ AI-generated insights and guidance
- ✓ Real-time compliance monitoring
- ✓ Historical analytics (Gold layer)

"Insights are generated in real time, not after the fact."

Databricks Services and Supporting Components Used

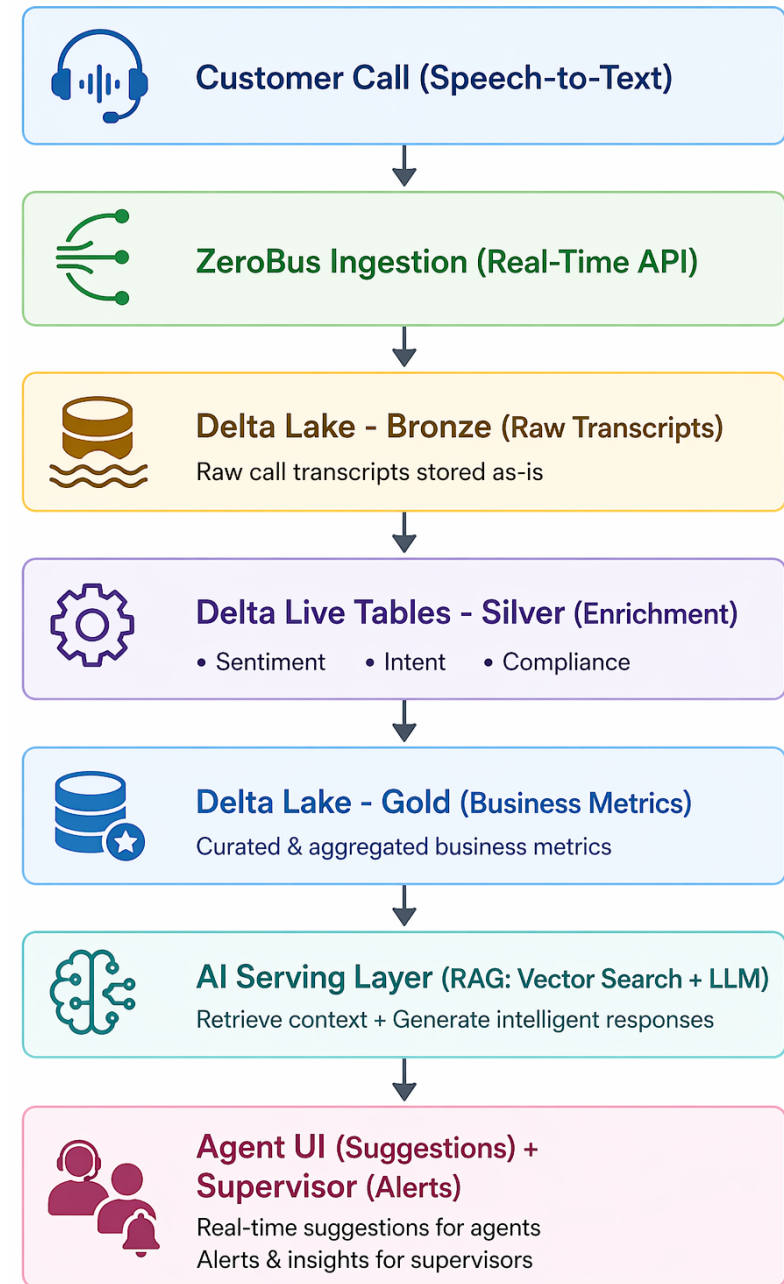
- **Delta Lake:** Stores raw, enriched, and aggregated call-center data across Bronze, Silver, and Gold layers with reliable table storage.
- **Delta Live Tables:** Processes incoming transcript records continuously and creates enriched Silver data, while scheduled DLT processing creates Gold analytics.
- **Unity Catalog:** Provides governance for tables and exposes UC Functions as controlled tools for the AI agent.
- **Databricks Workflows or Jobs:** Orchestrates scheduled DLT pipelines and batch post-call analytics processing.
- **LLM Integration:** Uses a Databricks-hosted LLM endpoint to generate real-time agent suggestions from live call context and retrieved knowledge.
- **RAG-style Knowledge Retrieval:** The agent retrieves relevant KB articles through `search_knowledge_base` before generating responses. Avoid saying confirmed Vector Search unless vector indexes are actually implemented.
- **Databricks SQL or Serving Layer:** Supports dashboard queries, analytics reporting, and downstream consumption from curated Delta tables.
- **Zerobus:** Streams transcript events into Databricks in near real time.
- **Speech-to-Text:** Converts call audio into transcript text before ingestion; the pasted content treats this as an upstream call-center capability.
- **Agent UI or Dashboard:** Displays live transcript, AI suggestions, member details, KB results, and compliance alerts for agents and supervisors.

High Level Architecture

End-to-End Flow: From Event to Insight

- Continuous data flow — events stream through Bronze → Silver → Gold layers without batch delays
- Data is enriched in real time — sentiment, intent, and compliance signals are derived in the Silver layer
- AI generates insights using enriched data and retrieved context (RAG: processed data + knowledge base)
- Users receive contextual insights and recommendations — agents get suggestions, supervisors get real-time alerts
- Continuous and near real-time processing — no batch delays, always current

"Simple, continuous, seamless — from event to insight."



ZeroBus

Simplified Real-Time Ingestion

- Direct-to-Lakehouse event ingestion — applications send events directly into Delta tables via APIs (gRPC/REST)
- Eliminates multi-hop complexity — no Kafka/Event Hub intermediaries, data flows directly from producers to Delta Lake
- Optimal for Lakehouse-first architectures — when Databricks is the primary destination, reduces latency and infrastructure overhead
- Sub-second data availability — events available for processing and AI inference within seconds of generation

Best suited when Databricks is the primary data platform

Zerobus Architecture

How Zerobus Ingest Works



Role of GenAI in The Architecture

How GenAI enables real-time contextual insights and assistance?

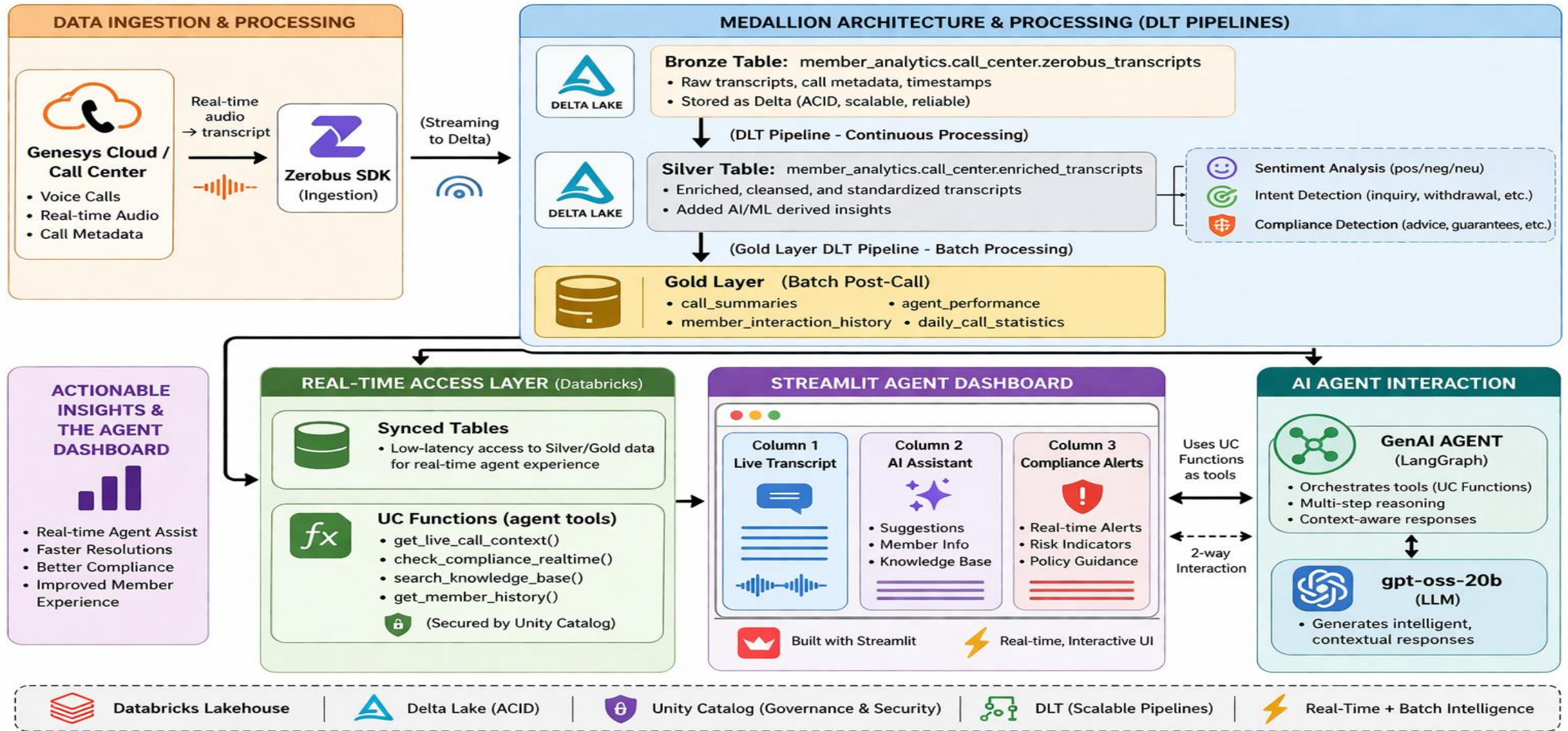
- GenAI processes natural language data from live transcripts to understand intent, sentiment, and context
- It combines enriched data (Silver layer) with retrieved knowledge (Vector Search) to generate relevant responses
- Instead of static rule-based systems, GenAI provides dynamic, context-aware recommendations
- Acts as a real-time assistant by suggesting responses, retrieving knowledge, and highlighting potential risks
- Shifts the system from post-event analysis to real-time insight generation

Traditional Systems vs. GenAI Systems

Traditional Systems		GenAI Systems
 Static keyword search Relies on exact keywords and predefined rules	vs.	 Contextual understanding Understands intent, context and conversation flow
 Manual retrieval Agents search multiple systems/documents manually	vs.	 Automatic suggestions AI instantly retrieves and recommends best answers
 Post-call analysis Insights available after the interaction	vs.	 Live assistance Real-time guidance and support during the call

"GenAI guides what happens next, not just what happened."

CALL FLOW ARCHITECTURE (Real-Time & Batch AI Pipeline)



Note: In this POC, we focus on Retrieval-Augmented Generation (RAG) for real-time insights. Model training, registration, and lifecycle management are not part of the current scope.

Bronze Layer – Raw Streaming Data Foundation

The Foundation: Raw Data as Source of Truth

- Captures incoming data exactly as received — real-time call transcripts are stored without modification
- Preserves raw events for traceability and reprocessing across downstream pipelines
- Enables consistent data foundation for both real-time processing and historical analysis
- Managed under Unity Catalog for secure and governed access

Bronze Layer Contents:

Delta Table: raw_call_transcripts

- timestamp
- call_id
- speaker (customer/agent)
- text (raw transcript)
- metadata (channel, agent_id)

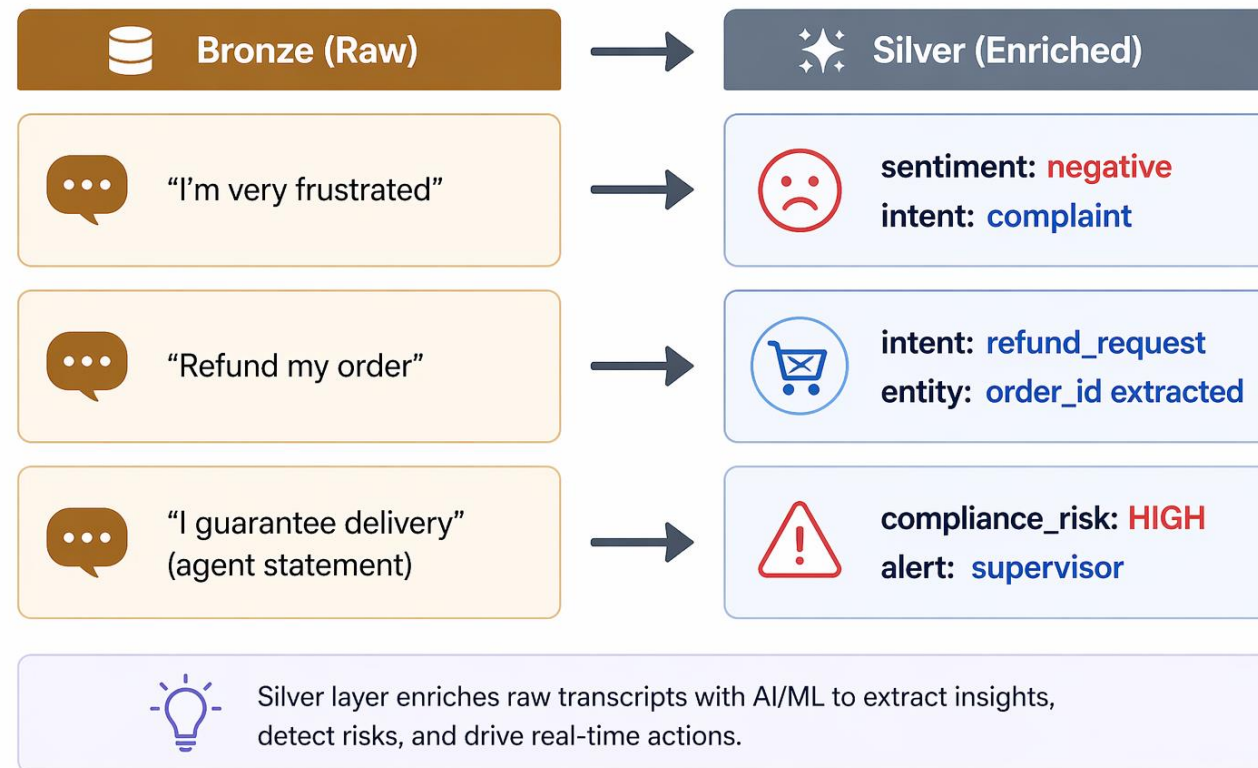
"Bronze layer ensures raw data integrity and enables reliable downstream processing."

Silver Layer - Real-Time AI Enrichment

From Raw Transcripts to Context-Aware Intelligence

- Real-time sentiment analysis — detects customer emotional state (positive / neutral / negative) to enable timely intervention
- Intent detection — identifies customer needs (refund, billing, support) to guide AI-assisted suggestions
- Entity extraction — captures key information (order_id, product, dates) to provide context for downstream processing
- Compliance signal detection — identifies potential risky language and surfaces alerts for review

Silver Layer Transformations



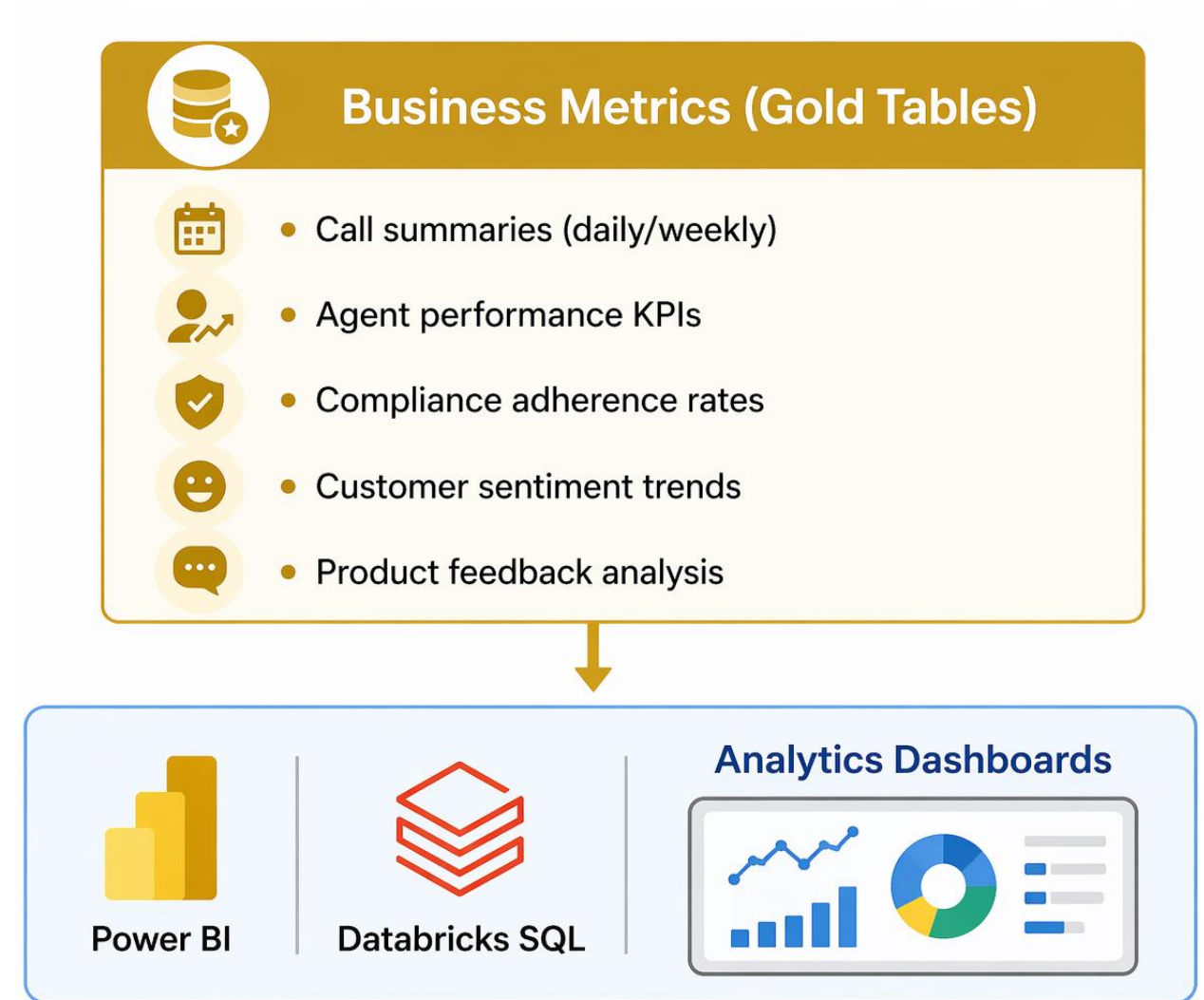
"Silver: transforms 'what was said' into 'what it means' — in real time."

Gold Layer – Aggregated Insights for Business Monitoring

From Real-Time Intelligence to Business-Level Metrics

- Aggregates enriched data from Silver layer into business-level insights
- Captures key metrics such as agent performance, sentiment trends, and interaction outcomes
- Enables trend analysis across calls, issues, and customer behavior over time
- Provides structured datasets for dashboards, reporting, and monitoring

"Gold layer converts real-time intelligence into measurable business insights."

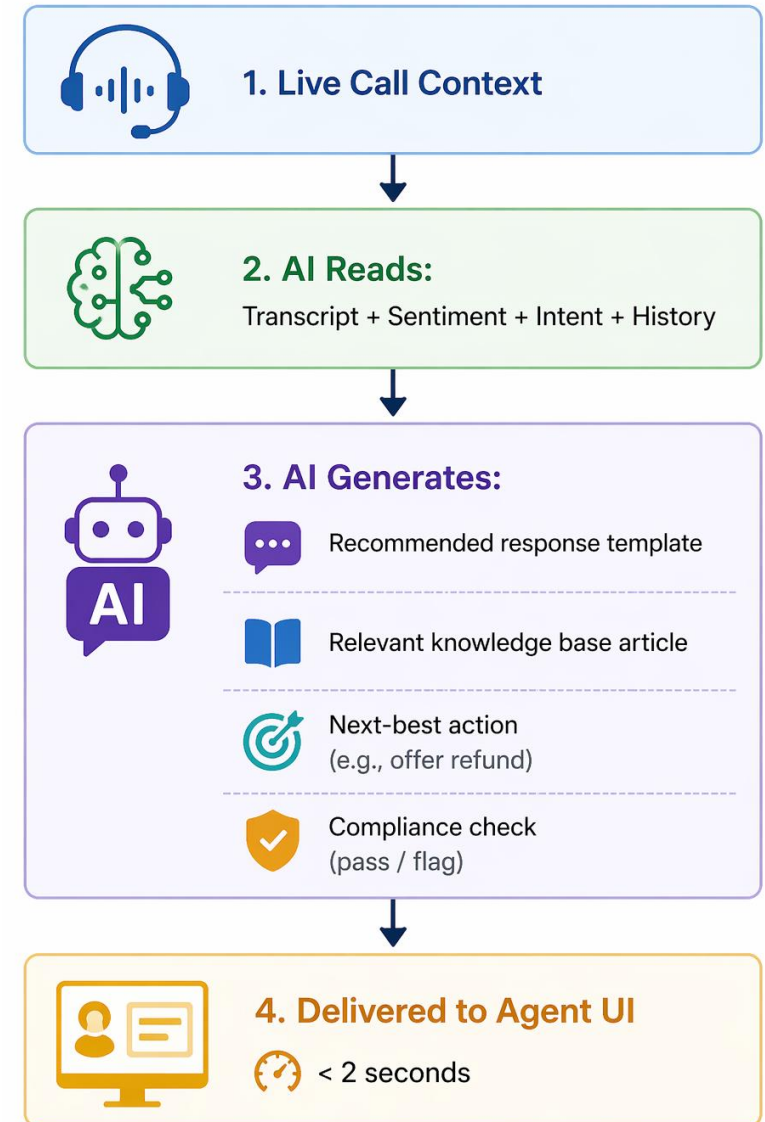


AI Integration – Context Aware Assistance in Real Time

Contextual Assistance During Live Interactions

- AI processes live conversation context — understands transcript, sentiment, intent, and available interaction context
- Suggests possible next actions — recommended responses, relevant knowledge articles, and guidance for handling the situation
- Reduces cognitive load on agents — eliminates need for manual search, provides instant contextual guidance
- Helps improve decision quality — enables faster, more informed responses and better customer experience

AI Integration Flow

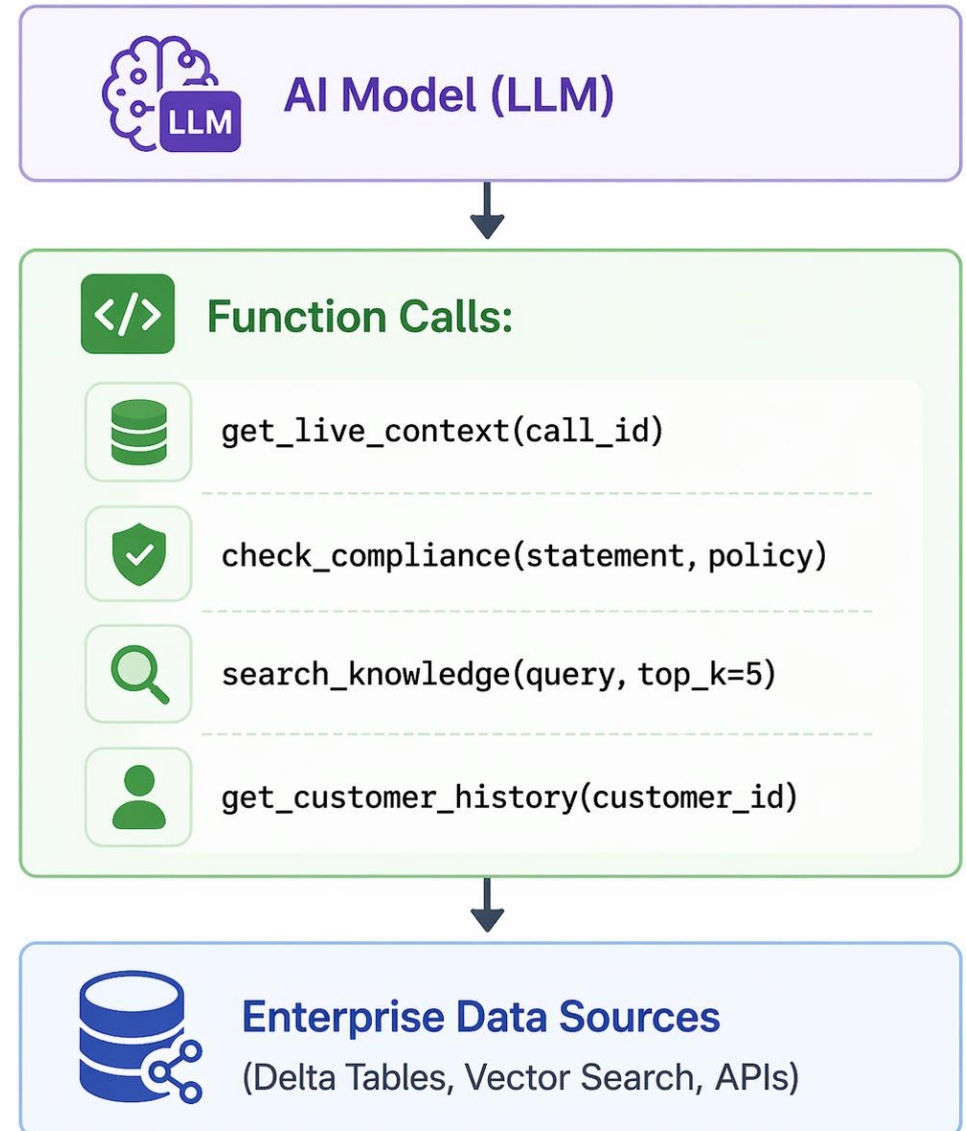


AI Tooling: Controlled Access to Enterprise Data

- Fetch live call context — retrieves current transcript, sentiment, intent, and interaction signals from processed data
- Check compliance rules — validates responses against policies, regulations, and business rules
- Search knowledge base — uses vector search to retrieve contextually relevant documents
- Retrieve customer history — accesses past interactions and preferences to enable personalized responses
- All access is mediated through governed functions — AI does not directly query raw data

“Functions provide controlled, secure access between AI models and enterprise data.”

Function Architecture

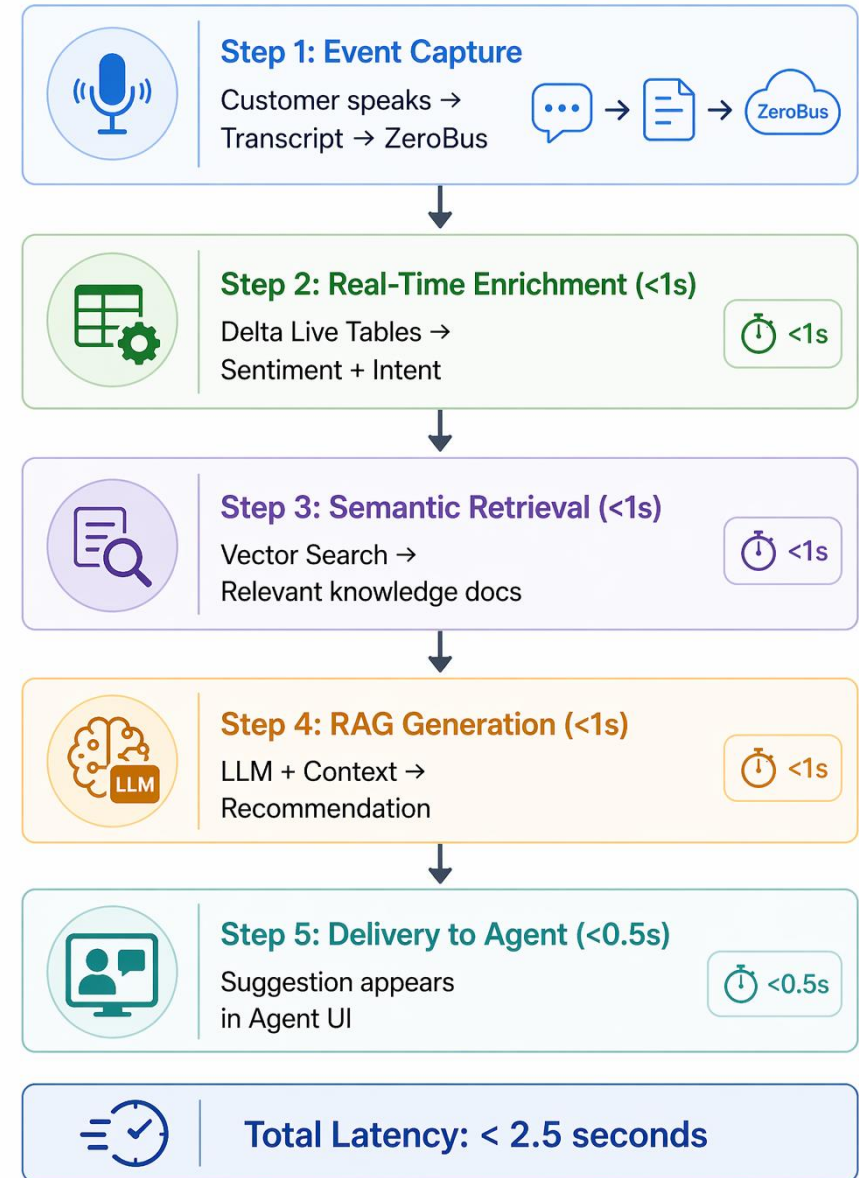


Real- Time Flow : From Event to Action

- Event captured — customer speech is converted to transcript and ingested into Databricks via Zerobus in near real-time
- Streaming enrichment — Delta Live Tables process incoming data continuously, extracting sentiment, intent, and compliance signals (Silver layer)
- Context retrieval — AI retrieves relevant knowledge using vector search combined with live conversation context (RAG pattern)
- Decision generation — LLM generates grounded next-best actions using retrieved context and structured signals
- Action delivery — recommendations and alerts are delivered to agent UI within seconds

“From customer conversation to AI-driven action — in near real-time.”

Complete End-to-End Flow



Business Value: Measurable Impact of Real-Time AI + Data

Measurable Outcomes

- Faster decision-making — reduces average handling time by enabling agents with real-time AI suggestions instead of manual lookup
- Improved customer experience — consistent, contextual responses based on live conversation + historical data
- Reduced compliance risk — real-time monitoring detects and flags violations during interaction, not after
- Lower operational complexity — unified platform reduces dependency on multiple tools and data duplication
- Expected improvements are based on industry benchmarks and POC simulation; actual impact depends on scale and adoption

Without Real-Time AI	Business Impact
 5-7 min avg handling	 ↑ Cost/call
 Post-call compliance	 ↑ Risk exposure
 5-7 disconnected tools	 ↑ Complexity

“Business value: faster decisions, better experience, lower risk, and simplified architecture — driven by real-time AI.”

Key Takeaways from the POC

- Real-time processing is essential — our POC demonstrates how streaming data enables decisions during the interaction, not after
- AI becomes operational — GenAI moves from generating insights to driving real-time actions using live context and knowledge retrieval
- Unified data + AI architecture — Delta Lake, streaming pipelines, and RAG simplify system design while improving performance and scalability
- Success metric shifts — value is measured by decision speed and accuracy, not just data volume or reports

Key Transformations

 From	To 
 Batch processing	Real-time streams 
 Static dashboards	AI assistants 
 Fragmented tools	Unified platform 
 Analyze past	Guide present 

“This POC demonstrates how data + AI systems move from observing events to driving real-time actions.”

Future

Roadmap: What Comes Next

- Advanced AI models — multi-agent systems, continuous learning from interactions, specialized domain models
- Enhanced search — hybrid Vector Search (semantic + keyword + metadata), multi-modal capabilities (text + voice + documents)
- Production integration — connect to CRM, ERP, order systems to enable AI to execute actions, not just recommend
- Enterprise-wide scale — extend architecture to sales, fraud detection, supply chain, predictive maintenance

thank you